

The Effect of Explainable AI and Uncertainty Quantification on Medical Students' Perspectives of Decision-Making AI: A Cancer Screening Case Study

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Abstract

Artificial intelligence (AI) offers potential to enhance healthcare decision-making but is limited by its 'black box' nature. Explainable AI (XAI) and Uncertainty Quantification (UQ) address these challenges by improving interpretability and reliability. Despite their potential, the impact of XAI and UQ on medical students' perception of AI in healthcare remains unclear. This study explores the impact of XAI and UQ on medical students' perceptions of AI in healthcare. A mixed-method study with 131 medical students from Singapore and China assessed the effects of varying AI methods on trust, usability, and decision-making. Results show that XAI and UQ enhance AI usability but highlight the need for clinically relevant explanations and contextualised uncertainty reasoning to optimise AI adoption in healthcare.

CCS Concepts

• Human-centered computing; • Human-computer interaction (HCI);

Keywords

Medical AI, Explainable AI, Uncertainty Quantification, Human-AI Interaction

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1 Introduction

Artificial Intelligence (AI) has rapidly emerged as a transformative force in healthcare, offering the potential to enhance diagnostic accuracy, optimize treatment plans, and personalized patient care [1]. For instance, machine learning algorithms have been employed to predict outcomes, medical imaging models have improved the analysis of X-rays, and natural language processing has streamlined the management of electronic health records [2–4]. These advancements aim to increase efficiency and support clinicians in making more informed decisions.

However, the integration of AI into clinical practice raises important questions about clinicians' autonomy and the nature of medical decision-making. Concerns emerge regarding whether AI might overshadow the clinician's role, leading to scenarios where humans uncritically accept machine-generated recommendations for critical decisions. This challenges the traditional model of patient care, where the clinician's judgement and expertise are paramount. Conversely, some clinicians may distrust AI to the extent that they disregard its input entirely, missing out on valuable insights that AI systems can provide. Balancing the benefits of AI with the preservation of clinician autonomy is essential to ensure that technology serves as a supportive tool rather than a replacement for human judgement [5]. These concerns are further compounded by the "black box" problem inherent in many AI systems. The lack of transparency and interpretability in AI makes it difficult for clinicians to understand how AI arrive at their conclusions [6]. Ethical and legal challenges, such as accountability in case of errors, further hinder the adoption of AI in clinical settings [7].

To address these issues, Explainable AI (XAI) has been developed to provide insights into how AI systems make decisions [8]. In healthcare, XAI can elucidate the rationale behind diagnostic suggestions or treatment recommendations, thereby enhancing transparency and trust. Methods such as saliency maps [9] and SHAP (SHapley Additive exPlanations) values [10] are examples

of XAI techniques. These tools allow clinicians to validate the AI's assessment against their own examination and clinical reasoning. Similarly, Uncertainty Quantification (UQ) measures the confidence levels of AI outputs, allowing clinicians to assess the reliability of AI-generated decisions [11]. UQ methods, such as Bayesian neural networks [12] and reconstruction uncertainty estimation [13–15], quantify the uncertainty inherent in AI predictions. UQ can alert clinicians when an AI's decision may be less certain, such as indicating a high degree of uncertainty in a diagnosis due to poor image quality or atypical patient data. UQ enables clinicians to make more informed decisions about when to trust AI recommendations and when to exercise additional caution, thereby promoting safer decision-making.

However, the implementation of XAI and UQ is not without challenges. Illogical or overly complex explanations provided by XAI can confuse users rather than clarify the AI's reasoning. For example, a “saliency map” might highlight irrelevant areas of an image, leading clinicians to question the validity of the AI's diagnosis. Similarly, high uncertainty scores from UQ can reduce confidence in AI recommendations, potentially causing clinicians to disregard useful AI input altogether. Understanding how these tools affect user emotions, trust, and the usability of AI systems is crucial for their effective integration into clinical practice.

Medical students represent the next generation of healthcare professionals who will interact with AI technologies. Their perceptions and acceptance of AI will significantly influence its future adoption in healthcare. By exploring their views, we can identify potential barriers and facilitators to AI integration and tailor educational programs accordingly. For instance, if medical students express concerns about AI undermining their diagnostic skills, curricula can be adapted to emphasise the complementary role of AI in supporting, not replacing, clinical judgement. This study aims to investigate how XAI and UQ affect medical students' perspectives on using AI in medical decision-making. Specifically, we explore:

- User Emotions: How do XAI and UQ influence medical students' emotional responses, such as confidence or anxiety, when interacting with AI systems?
- User Trust: What impact do XAI and UQ have on the trust that medical students place in AI-assisted decisions?
- Usability: How do XAI and UQ affect the perceived ease of use and usefulness of AI tools in clinical scenarios?

We aim to provide insights into the benefits and limitations of XAI and UQ from the users' perspective. Our findings indicate that while XAI and UQ can indeed enhance the usability of AI systems by facilitating data visualisation and assessing decision reliability, they may also decrease user faith due to instances of illogical explanations and high uncertainty scores. These results highlight the need for developing clinically sound and relevant explanations and contextualised reasoning within AI systems. Our study also underscores the importance of incorporating AI education into medical curricula to better prepare future clinicians for the challenges and opportunities that AI presents in healthcare.

2 Literature review

Medical students' opinion to AI has been studied in several papers. It is generally agreed upon that most medical students perceive

themselves as having at least a basic understanding of AI principles [16, 17]. Most also harbour a positive attitude towards AI [17–19] and think that it should be incorporated into the medical education curriculum [17, 18]. Students felt that AI can benefit healthcare by speeding up processes [16], reducing medical errors and improving medical decision accuracy [20, 21]. A concern that one study found is that AI will take over the role of doctors and threaten the job security of doctors [22]. However, other studies found that the majority of students surveyed disagreed with this sentiment and thought that AI will be used as an assistive technology [20], rather than take the place of doctors, such as radiologists or dermatologists [21], since AI would not be able to establish definitive diagnosis on its own [18]. Another fear is that the practice of medicine may become less humane if the role of AI begins to overshadow that of the human clinician [20].

AI can be applied in healthcare in many ways. Some examples include making diagnostic decisions for illnesses such as cancer [23], identifying suspicious areas of concern on imaging or other investigations [24, 25], and recommending treatment. In terms of performance, multiple AI models have been shown to be equally capable, or sometimes better than human clinicians who are less experienced [26, 27]. Yet, despite many studies proving its success, user trust in AI remains a major barrier to the adoption of AI in clinical practice [28]. This is due to the “black box” nature of AI – users often have a poor understanding of how AI arrives at its decision. Without knowing the rationale behind each decision, users are less trustful of AI and are reluctant to rely on it [29, 30]. This issue is even more prominent in healthcare since medical decisions can have a significant impact on people's lives. Thus, AI has to be more explainable and align with current medical practices, guidelines and regulations in order for clinicians to trust it [31]. XAI aims to address this problem of user trust by providing explanations that increase interpretability and thus transparency [32]. Interpretable AI helps clinicians assess the degree of agreement between the AI and the clinician's own judgement, allowing more informed decisions.

Some examples of XAI being used in healthcare include diagnostic decision-making, treatment planning and monitoring devices that provide warnings when an adverse event may happen. For example, XAI has been applied in explaining early detection of acute critical illness by picking out relevant parameters from the patient's health records, allowing clinicians to understand why the AI predicted the patient to be in a critical state [33]. Some XAI models are also able to translate the critical information into sentences that are using appropriate medical vocabulary. For instance, XAI has been used to explain why lung nodules found on CT scans are suspicious by describing its appearance according to features such as shape and contour [34]. An example of XAI used in treatment planning is the recommendation of treatment options for breast cancer patients. The XAI model outlines critical patient features and recommend the best treatment option [35]. Studies have found that the use of XAI can increase confidence levels in AI. A 2024 study found that by increasing transparency, XAI improved dermatologists' confidence in their diagnoses and their trust compared to conventional AI without explanations [28]. XAI has also shown to improve confidence in non-experts in fields like histopathology [36].

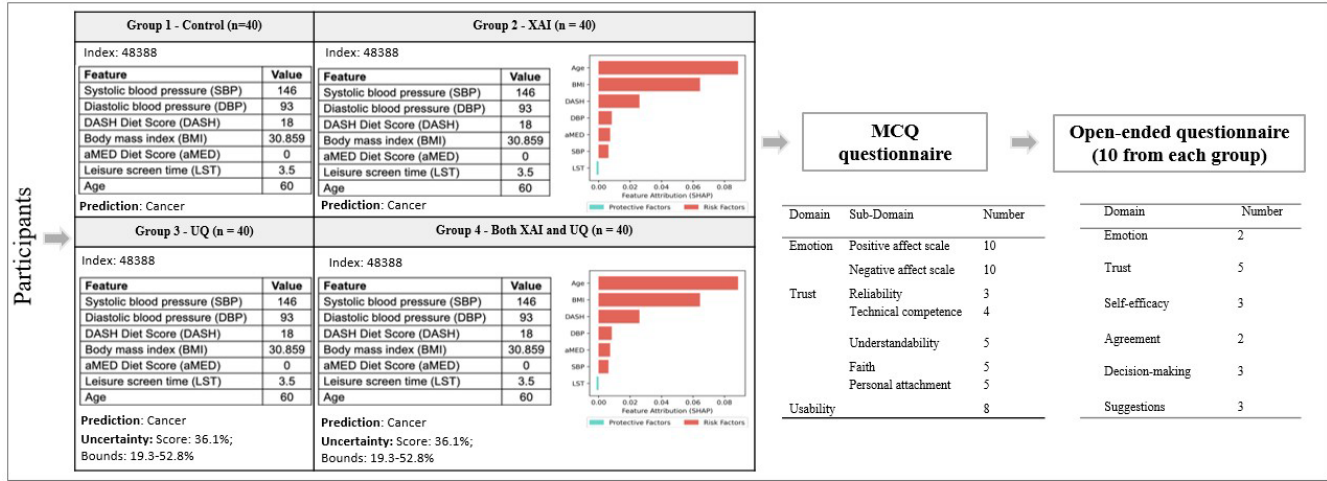


Figure 1: Study design and examples of case studies were provided to participants in each group. This study recruits 160 medical students at both undergraduate and postgraduate levels from medical schools in Singapore and China. Students are randomly assigned to one of four groups, representing four comparison conditions. A MCQ Questionnaire covering three study domains, along with an open-ended questionnaire addressing six study domains, are used to collect their feedback.

UQ is a technique enhancing the use of AI in medicine [37]. Uncertainty score tells a clinician which cases are likely to be straightforward, versus cases which are ambiguous and require manual checking. This is known as segmentation and uncertainty-based referral which helps users determine how much confidence to place in an AI's decision [38]. Some of the more common estimation methods used to derive UQ for medical imaging include Monte-Carlo Dropout and Deep Ensemble [39], which have been applied to the detection and classification of diseases from fundus, histology, dermoscopic, MRI and X-ray images [38, 40, 42, 43]. It has been proposed that UQ should be complemented with explanations [41].

3 Methods

A mixed-method study [44] was conducted on a total of 160 medical students from Singapore and China. Participants were randomly split into four groups. Each group was shown 4 cases of colon cancer predictions and 4 cases of lung cancer predictions made by AI models respectively, together with the features used by the AI model for prediction. Detailed discussion on the AI models, training datasets, XAI and UQ are presented in the appendix (A.1). Participants were also given information on the dataset on which the models were trained and the overall accuracy of each model. Additionally, the second group received a bar chart displaying XAI feature attribution scores, the third group received UQ, and the fourth group received both (Figure 1). The participants then filled up a Google form containing multiple-choice questions. Those interested were invited to share their thoughts through an open-ended questionnaire (Figure 1). Participants who did not complete the survey were excluded from the study.

The Multiple Choice Question (MCQ) questionnaire explored user emotions, user trust and usability with the **Positive and Negative Affect Schedule** [45], **Human-Computer Trust (HCT) System** [46] and **System Usability Scale (SUS)** [47] respectively.

To cater to the study context, some questions from the HCT system and SUS were removed or edited. The final list of questions for the questionnaire can be found in A.4. In the open-ended questionnaire, participants were given open questions to explain their emotions, degree of trust in the AI models, how self-efficacious they felt the models were, their agreement with the models' results, the models' impact on their decision-making, and future uses for the AI models. The MCQ questionnaire used a Likert scale and the scores for **each domain were summed up**. Each dataset was tested for normality using Shapiro-Wilks test and homogeneity of variance using Levene's test. When the assumptions of homogeneity and normality were met, Analysis of variance (ANOVA) [21] was used; otherwise Kruskal-Wallis test was used. The responses for the open-ended questionnaire were analysed using inductive thematic analysis [22].

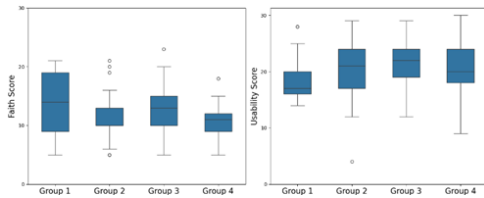
4 Results

A total of 131 participants completed the MCQ questionnaire (their demographics are shown in A.2.) The number of responses gathered for each group was 37, 33, 33 and 28 respectively. We found statistically significant differences in faith [p-value = 0.0460] and usability [p-value <0.001] across the four groups. All other differences across groups and across expertise levels were found to be statistically insignificant (Table 1). Interestingly, when provided with the bar chart produced by XAI and/or uncertainty scores, students reported having less faith in the AI models when they were unsure (Figure 2). After being given the bar chart produced by XAI and/or uncertainty scores, students reported that the AI model was more usable (Figure 2).

A total of 46 participants completed the open-ended questionnaire (their demographics are shown in A.3.). The number of responses collected from each group was 11, 12, 13, and 10, respectively. We first examine two dimensions—usability of AI systems and faith in AI—where statistically significant differences were

Table 1: Mean and P-values for each subdomain.

Domain		Mean				P-value
		Group 1	Group 2	Group 3	Group 4	
Emotions	Positive affect score /	23.11	23.73	27.33	23.43	0.122
	Negative affect score (max=50)	15.43	13.52	13.97	12.54	0.245
Trust	Reliability (max=15)	6.86	8.24	7.94	8.11	0.057
	Technical competence (max=20)	13.84	13.45	13.76	12.57	0.247
	Understandability (max=25)	18.00	18.03	17.48	17.96	0.453
	Faith (max=25)	13.51	11.21	12.82	10.46	0.046
	Personal attachment (max=25)	15.16	13.55	14.70	12.96	0.135
	Overall HCT (max=105)	67.38	64.48	66.70	62.07	0.331
Usability (max=32)		18.22	20.45	21.11	20.71	<0.001

**Figure 2: Box and whisker plots of faith and usability across groups**

identified among the four participant groups. Next, we provide a summary of additional insights gathered from the open-ended responses regarding AI decision-making in general. The results are presented as follows.

4.1 How did XAI and UQ increase the usability of AI?

(1) XAI increased the visualisation of data and provides an overview of the patient’s status. The XAI made use of colours to represent risk vs protective factors which allowed the students to quickly associate features which were protective factors versus those which were risk factors. Students were able to obtain a quick overview of the patient’s status and know which features to focus on.

“...The risk factors and protective factors were depicted in graphs showing how strongly positive or negative risks that the particular patient had. This lets me compare the bar graphs along with the clinical history taken, in order to learn what kind of health the machine rates as a big green bar compared to a small green bar.”

(2) UQ helped students decide when to use AI. Students quickly noted that straightforward cases tended to have low uncertainty scores while cases with more opposing factors had higher uncertainty scores. The scores gave them a quantifiable measure of how confident the model was in its predictions and the model’s accuracy for each case and they quickly knew which cases they could trust, versus cases. Thus, students easily knew which cases they could trust the AI in and diverted their attention towards cases where the uncertainty was higher.

“...I feel like I know when to trust the AI more [when it had a lower uncertainty score] and when to rely more on my own clinical reasoning when the AI had a higher uncertainty score.”

4.2 How did XAI and uncertainty scores decrease students’ faith in AI?

Students exhibit decreased confidence in AI systems with XAI and UQ. We believe the root issue lies in the lack of understanding of how XAI generates its explanations and how the uncertainty values are determined.

(1) XAI results can be inconsistent with medical students’ knowledge. There were certain feature attributions that students disagreed with based on their own clinical knowledge. Students thought that some features should not have been used at all, while others should have a different weightage. As a result, some of the feature attributions did not appear to be based on sound clinical reasoning to some students.

“The part of the model’s results I disagree with is the protective factors as not all of them are protective factors per se and should not necessarily contribute to the decision-making process if the values are not high enough.”

(2) Difficulty understanding how XAI determined feature attributions. Students were also confused over how XAI used SHAP to determine the feature attributions for each factor, such as whether they were protective or risk factors, and the weightage of each feature attribution relative to each other.

“I do not understand the AI results, because even if the conclusion aligns with my own, the amount of importance/weight placed on each factor was unclear, as well as the interplay of different factors.”

XAI algorithms, such as SHAP, provide “individualised explanations,” meaning that the same feature can have varying impacts on the prediction outcome. However, participants did not fully understand this concept. There were instances where SHAP assigned completely different attributions to the same feature, which students found difficult to comprehend.

“...the fact that the same amount of screen time [3.5h] is considered a risk factor in one patient and a protective factor in another does not make sense to me.”

(3) High uncertainty score and wide uncertainty bounds. Students were distrustful of the predictions when they perceived the uncertainty score as too high. When uncertainty bounds were wide, some students found it difficult to interpret the AI's confidence. The resulting confusion made students concerned about how much they could rely on the uncertainty score as a measure of confidence in the AI's accuracy, given that the uncertainty could potentially be very high.

"I felt hesitant because I realised that there was actually quite a large margin for error in determining how useful the model would be. In addition, it was difficult to gauge the uncertainty levels - bringing about a problem about how it was difficult for an individual to process raw statistics."

(4) UQ does not provide more insight beyond a statistical probability. Students felt it would be more helpful if they understood how scores were calculated. Providing a context to the uncertainty could help users identify which areas of the patient's profile are most controversial and may guide them towards relevant investigations.

"It would be more helpful if the uncertainty score could be accompanied by a more detailed explanation of what factors contribute to high or low uncertainty... This additional context would empower clinicians to make more informed decisions... Additionally, integrating visual aids or explanations that break down the sources of uncertainty could make the AI's predictions more transparent and easier to interpret..."

4.3 Sentiments on AI supporting medical decision-making

(1) AI methods improve decision-making in general. AI Decision-support with XAI and UQ can act as a second opinion for clinicians to check their decisions against. It may help identify cases that clinicians may miss, and prompt clinicians to rethink a case when the AI's decision is incongruent with the clinician's own decision.

"I felt more confident in my clinical judgement regarding the patient's risk of cancer when I used the model because I am able to cross reference with a 2nd party. . . and rethink my decisions accordingly."

Moreover, in ambiguous cases or situations where users lack confidence, AI can offer an initial reference point for clinicians to build upon. Even if the AI's recommendations are not flawless, they can still guide clinical reasoning by narrowing down possibilities or highlighting relevant factors. This support allows clinicians to make more informed decisions by using AI as a complementary tool, rather than relying on it as the sole source of judgement.

"The AI method impacted my decision in terms of giving me an answer that has ~80% chance of being correct when I myself had no answer/ was very uncertain about my conclusion at 0% confidence."

Lastly, AI is able to analyse large amounts of data quickly. In the case of medical decision support, by providing clinicians with a quick overview of each patient's case and highlighting important factors, AI can speed up processes and help clinicians prioritise cases that require more attention when workload is high.

"...as a doctor, this might be quite helpful to estimate patient outcomes, and take into account all the relevant lifestyle and past medical history. I would appreciate using this tool alongside my clinical judgement. . ."

(2) There are several risks identified in using medical AI. Firstly, there were concerns about medicolegal and ethical issues in the case of a missed diagnosis or misdiagnosis. If the AI had incorrectly predicted an outcome, the legal implications are unclear – is the AI or the clinician who trusted it at fault? Without sound clinical reasoning, students were not keen on relying on AI for diagnosing serious conditions if its accuracy was not 100%.

"...some of the . . inaccurate predictions make me a bit wary of relying entirely on this AI model. . . we are talking about human lives and I am not very comfortable using this model entirely when the stakes are so high."

Secondly, due to the lack of medical education on AI, some participants felt that unfamiliarity with AI could counterintuitively hinder productivity. In addition to understanding the medical case, users must also learn to interpret the AI's outputs. This challenge is further exacerbated by the complexity of existing AI, XAI, or UQ algorithms.

"...limitations the uncertainty score may face. . . distrust from providers and patients alike, as they may not understand the tool well. The pattern recognition of AI may also eventually uncover patterns that are unexplainable to us logically but only understood by the model itself, forming results that are bizarre yet correct."

Lastly, AI may lead clinicians to rely on it rather than fully analysing cases themselves, diminishing their critical thinking and decision-making skills. This issue is concerning when the AI selectively prioritises factors it deems relevant to its predictions. As a result, two conflicting problems arise: clinicians may overlook the factors the AI model considers in its calculations, while the AI, in turn, cannot incorporate other important information outside its training data. Therefore, overreliance on imperfect AI models without a clear understanding of their limitations poses significant risks.

"... healthcare professionals could place too much trust in the AI's output without fully considering the clinical context, leading to overconfidence in the AI's decisions. This could be particularly problematic in situations where the uncertainty score is low, but the AI's prediction is incorrect due to unseen biases . . ."

5 Discussion and limitations

From diagnosing diseases such as cancer [23] to identifying suspicious areas of concern on medical images [24, 25], and recommending personalised treatment options [35], AI has proved to be useful in healthcare in many ways. While AI has demonstrated great potential in enhancing diagnostic accuracy and efficiency [48], this study explored medical students' perceptions of the impact of XAI and UQ on their decision-making in clinical contexts. Understanding how these tools influence emotions, trust, and usability is essential for their successful integration into clinical practice.

Our findings indicate that XAI and UQ can increase the usability of AI. XAI may reduce cognitive load by transforming complex data into visual representations, such as bar charts, thereby simplifying information processing for students. Cognitive load theory suggests that reducing unnecessary information helps free up processing resources and working memory [49]. XAI highlights key diagnostic features, allowing students to focus on relevant factors and avoiding being distracted by useless information. UQ provides students a quantitative reference point that facilitates informed decisions.

However, this increased usability poses potential risks. Students may become overly reliant on AI's decisions and ignore the limitations of AI, particularly in situations with low uncertainty scores. Furthermore, there are questions about how AI may shift responsibilities in the healthcare system. Concerns have been raised that physicians may become overly dependent on AI, which could lead to a gradual decline in their clinical skills and connections with patients [50]. In turn, AI developers are responsible for ensuring safe, useful systems and positively influencing public health perceptions [50]. It is crucial to develop medical students' AI literacy by introducing AI courses in medical schools, thereby reducing the knowledge gap between AI and medicine. This is consistent with several studies [51–53], which emphasised the importance of integrating AI training into medical education. With training, medical students cultivate the skills to critically evaluate and use AI in clinical settings [54], ensuring informed decision-making and mitigating the risks of over-dependence.

Previous research suggests that XAI can enhance trust by offering transparent explanations of AI decisions [55]. However, our findings, consistent with other studies, reveal that the interpretations extracted by XAI do not always improve trust [56] or align with physicians' expectations [57]. Our study found that when the explanations provided by XAI contradict students' prior knowledge, it can instead reduce their faith. This issue is especially pronounced in cancer diagnosis, where the high stakes make physicians approach AI predictions with increased caution. Additionally, difficulties in comprehending how XAI determines feature attributions were noted. This suggests that medical education should incorporate AI training, thereby better equipping future clinicians to use and evaluate AI systems within clinical contexts.

In light of our results, we believe that incorporating UQ in AI methods for healthcare reduces the likelihood of overreliance on AI. The decreased faith in the AI's predictions in cases with high uncertainty show that when users are told that the model is less certain of its own prediction, they heed this piece of information as a warning and choose to avoid relying on the model's predictions for such cases. Based on the open-ended responses gathered, we see that most participants are presently aware and wary of possible over-reliance on AI. Most also concluded that they would reserve the use of AI as a second opinion to enrich their own thought processes, rather than use it as the sole tool.

Our study has some constraints. First, the sample size was modest, with 131 out of 160 participants completing the study. Additionally, the study was limited to Asian participants. Moreover, students were exposed to a specific subset of AI—decision-making AI for cancer diagnosis. As a result, the findings may be most relevant to similar AI applications and may not fully generalise to other healthcare AI tools or decision-making AI designed for

different conditions or purposes. Future research should expand the study to include participants from diverse regions and professional backgrounds, which may enhance its relevance and applicability. Additionally, future work can investigate how XAI and UQ influence clinician autonomy and accountability in real-world practices. Furthermore, exploring the long-term impact of integrating XAI and UQ into medical education would provide valuable insights into optimizing AI tools for clinical practice.

6 Conclusion and Recommendation

This mixed-method user study explored the impact of XAI and UQ on medical students' perceptions towards AI's role in clinical decision-making. The background issue hindering the adoption of AI in healthcare is the "black box" nature of AI leading to hesitancy among medical professionals in trusting AI decisions. XAI aims to make AI outputs more interpretable by explaining the decision-making process, while UQ quantifies how reliable these decisions are. Our main findings include that XAI and UQ enhanced the usability of AI tools by providing visual aids and measures of reliability, but decreased faith in AI tools. This was primarily due to sometimes confusing explanations and high uncertainty scores, which led to scepticism in the AI's ability to make accurate and relevant decisions. In conclusion, while XAI and UQ offer potential in improving the usability of AI in healthcare, they also present challenges. The findings suggest a need for more clinically sound explanations by XAI and contextualised reasoning behind to adequately induce trust in AI methods. Furthermore, incorporating AI literacy into medical education could help future clinicians effectively utilise these technologies without over-relying on them.

Based on our results, we propose a few theoretical and practical areas for future AI development in healthcare. Firstly, we propose that UQ methods be complemented with an explanation behind its output. This will provide users with meaningful insights into sources of uncertainty beyond just a statistical probability. Secondly, we found that diagrams were useful in presenting information. Future XAI models should use visual explanations to increase its understandability. Lastly, students with less background knowledge in AI were often confused, distrustful and sometimes wrongly interpreted outputs. Hence, we call for the introduction of medical education on AI to address potential widespread use of AI in the future.

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Appendix

A.1 Detailed information on the XAI and UQ models used.

Dataset

This study uses two medical datasets – Lung and Colorectal cancer datasets. Both were derived from the Singapore Chinese Health Study (SCHS) [1] which studies the effects of diet, environment and genetics on the development of cancer and other chronic disease in the ethnic Chinese population in Singapore.

The Lung cancer dataset consists of 23622 instances. Each instance describes the condition of a patient using 5 categorical and 16 continuous features and a class label describing if the patient has Lung cancer. The Colorectal cancer dataset consists of 22901 instances. Each instance describes the condition of a patient using 5 categorical and 12 continuous features and a class label describing if the patient has Colorectal cancer.

Both datasets were pre-processed with the following steps: Patients with missing values were dropped, continuous features were min-max normalised while categorical features were one-hot encoded. We randomly divided the dataset into training (80%), validation (10%) and test sets (10%) for model training, tuning and evaluation respectively. As both datasets are class-imbalanced, we oversampled the cancer instances in the training set with SMOTE [2].

We selected four instances from the test sets of each dataset as examples for the survey. These four instances represent all combinations of the class labels (Normal and Cancer) and correct or incorrect predictions.

Table 2: Description of Lung and Colorectal cancer datasets and model parameters for each model.

Data Subset	Cancer Proportion (%)	Size		Feature Selection		Hyperparameters	
		Before Dropping	After Dropping	Features	Selected Features	Trees	Max Depth
Lung	1.90	23622	17464	21	15	50	10
Colorectal	2.58	22901	20689	17	7	50	5
Lung	1.90	23622	17464	21	15	50	10

A.1.1 AI Model

We trained a Random Forest [3] model on each dataset. Random Forest is an ensemble method which aggregates the predictions of multiple decision trees improving its accuracy and robustness to overfitting, which has made it a widely studied model for medical diagnosis [4, 5]. Each decision tree predicts the class of a new instance by sorting the instance into a leaf node based on its features. The majority class in the leaf node determines the class prediction for that instance. To make the final prediction, the random forest model aggregates predictions from each decision tree through majority voting.

To improve prediction performance, we employed feature selection. We estimated the mutual information [6] between each feature and the class label. A high mutual information indicates that the feature is important for prediction. We incrementally added features to the feature set in descending order of importance and evaluated models trained on the feature set on the validation set using AUC (Receiver Operating Characteristic Area Under the Curve). We selected the feature set with the highest AUC. To determine the hyperparameters for the random forest model (Number of trees and max depth), we used grid search $n_trees\{50, 100, 200, 300, 400, 500, 600, 700\}$ and $max_depth\{5, 10, 20, 30, 40\}$ and selected the hyperparameters with the highest AUC on the validation set. Table 2 indicates the number of selected features and hyperparameters for each dataset. Table 3 displays the performance of each model on each data subset.

Table 3: Model prediction performance.

Data Subset	Lung Cancer		Colorectal Cancer	
	AUC	Accuracy	AUC	Accuracy
Train	0.968	0.829	0.999	0.862
Valid	0.816	0.818	0.623	0.833
Test	0.734	0.802	0.632	0.831

A.1.2 XAI Method

We utilised SHAP [7] (SHapley Additive exPlanations), a widely used XAI technique in medical diagnosis [8], to derive feature attribution scores from each model for each instance in the survey. Each instance has as many SHAP feature attribution scores as the number of features in the dataset. The sign of the score indicates how the feature affects prediction – negative values indicate that the feature negatively

contributes to the prediction and vice versa. The magnitude of the score indicates how strongly the feature attributes prediction – high feature attribution values indicate that the feature is important for prediction. To improve comprehensibility, we have displayed the feature attribution score on a bar chart in the survey and indicated risk factors in red and protective factors in green together with a legend.

A.1.3 UQ Method

Besides providing respondents with the overall accuracy of each model (Indirect Uncertainty), we also provide uncertainty score and bounds (Direct Uncertainty) [9] derived from the Entropy [10, 11] of prediction probabilities. Entropy is the most accessible uncertainty estimate for classification problems. It is computed as the Shannon's entropy (Equation 1) of ML prediction probabilities $p(x)$.

$$h = -p(x) \log p(x) \quad (1)$$

To improve the interpretability of our uncertainty score, we further processed Entropy to yield an uncertainty percentage. To derive this value, we first define incorrect prediction probability e as the absolute difference between the prediction label (y ; 1 for cancer, 0 for normal) and prediction probability y .

$$e = |y - y| \quad (2)$$

Next, we fit a multivariate Gaussian distribution on the entropy h and incorrect prediction probabilities e of the validation set. With this multivariate Gaussian distribution, we can compute the conditional mean and variance of incorrect prediction probability e given the entropy of this new instance [12], $\mu(e|h)$ and $\sigma^2(e|h)$. Since prediction probabilities are in the range of 0 and 1, we further truncate [13] the conditional Gaussian distribution such that $x \in [0, 1]$, yielding a modified conditional mean $\mu(e|h)$ and variance $\sigma^2(e|h)$. The modified conditional mean $\mu(e|h)$ serves as the uncertainty probability of the instance. We also compute the lower and upper bound of uncertainty by adding and subtracting $1.96 \times \sigma(e|h)$. This yields an uncertainty bound with a 0.95 probability that the true uncertainty of the model falls within these bounds.

Table 4 evaluates the performance of our uncertainty score on each data subset. We observe a high correlation between incorrect prediction probabilities and our uncertainty score in both datasets.

Table 4: Uncertainty quantification performance.

Data Subset	Lung Cancer	Colorectal Cancer
Train	0.755	0.845
Valid	0.745	0.780
Test	0.713	0.775

A.1.3 References

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A.2 Participants in the MCQ study.

Group	Group 1	Group 2	Group 3	Group 4	Total
Clinical	23	19	21	19	82
Preclinical	14	14	12	9	49
Total	37	33	33	28	131

Group	Group 1	Group 2	Group 3	Group 4	Total
Singapore	33	30	30	24	117
China	4	3	3	4	14
Total	37	33	33	28	131

A.3 Participants in the open-ended questionnaire study.

Group	Group 1	Group 2	Group 3	Group 4	Total
Clinical	3	9	8	7	27
Preclinical	8	3	5	3	19
Total	11	12	13	10	46

Group	Group 1	Group 2	Group 3	Group 4	Total
Singapore	9	10	12	9	40
China	2	2	1	1	6
Total	11	12	13	10	46

A.4 Final list of questions for the MCQ questionnaire and open-ended questionnaire.

A.4.1 Emotion – Positive and Negative Affect Schedule.

Indicate the extent you have felt this way.

Interested; Distressed; Excited; Upset; Strong; Guilty; Scared; Hostile; Enthusiastic; Proud; Irritable; Alert; Ashamed; Inspired; Nervous; Determined; Attentive; Jittery; Active; Afraid

Likert scale used:

Point	Description
1	Very slightly or not at all
2	A little
3	Moderately
4	Quite a bit
5	Extremely

Positive Affect Score: Add the scores on items 1, 3, 5, 9, 10, 12, 14, 16, 17, and 19. Scores can range from 10 – 50, with higher scores representing higher levels of positive affect.

Negative Affect Score: Add the scores on items 2, 4, 6, 7, 8, 11, 13, 15, 18, and 20. Scores can range from 10 – 50, with lower scores representing lower levels of negative affect.

A.4.2 Trust – Human-Computer Trust.

Please indicate how strongly you agree or disagree with the following statement:

Strongly disagree to Strongly agree: 1-5

Category	Sub-category	Question
Trust	Perceived Reliability	R1 - The model always provides the advice I require to decide whether a patient is likely to have cancer or not. R2 - The model performs reliably. R3 - I can rely on the model to function properly.
	Perceived Technical Competence	T1 - The model uses appropriate methods to reach decisions. T2 - The model has sound knowledge about cancer prediction built into it. T3 - The advice the model produces is as good as that which a highly competent person could produce. T4 - The model makes use of all the knowledge and information available to it to produce its solution to the problem.
	Perceived Understandability	U1 - I know what will happen the next time I use the model because I understand how it behaves. U2 - I understand how the model will assist me in deciding whether a patient is likely to have cancer or not. U3 - Although I may not know exactly how the model works, I know how to use it to make decisions about the problem. U4 - It is easy to follow what the model does.
	Faith	U5 - I recognize what I should do to get the advice I need from the model the next time I use it. F1 - I believe advice from the model even when I don't know for certain that it is correct. F2 - When I am uncertain about a decision I believe the model rather than myself. F3 - If I am not sure about a decision, I have faith that the model will provide the best solution. F4 - When the model gives unusual advice I am confident that the advice is correct. F5 - Even if I have no reason to expect the model will be able to solve a difficult problem, I still feel certain that it will.
	Personal Attachment	P1 - I would feel a sense of loss if the model was unavailable and I could no longer use it. P2 - I feel a sense of attachment to using the model. P3 - I find the model suitable to my style of decision making. P4 - I like using the model for decision making. P5 - I have a personal preference for making decisions with the model.

A.4.3 Usability – System usability scale.

Please indicate how strongly you agree or disagree with the following statement:
Strongly disagree to Strongly agree: 1-5

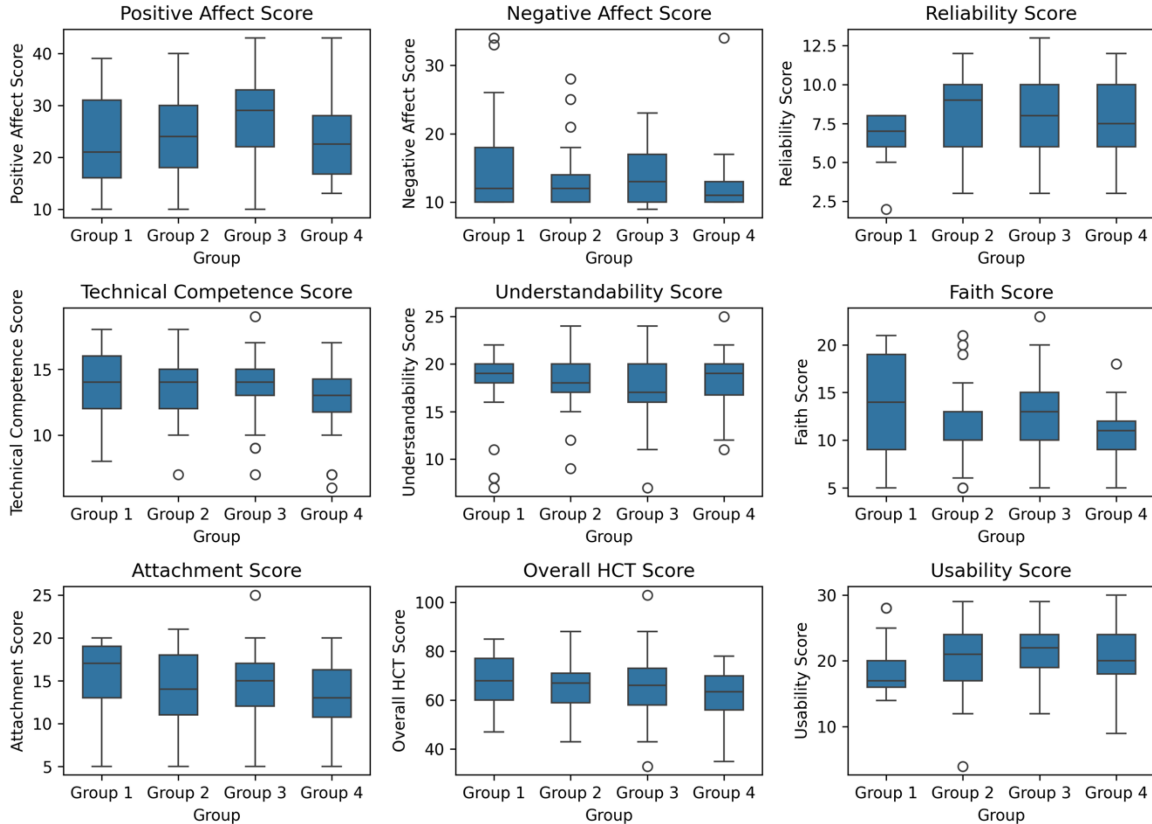
1. I think that I would like to use this model frequently.
2. I found the model unnecessarily complex.
3. I thought the model was easy to use.
4. I think I would need the support of a technical person to use this model.
5. I would imagine that most people would learn to use this model very quickly.
6. I found the model very cumbersome to use.
7. I felt very confident in my ability to using the model.
8. I needed to learn a lot of things before I could get going with this model.

A.4.4 Agreement:

I agree/disagree with the XX result of the AI model.
Open-ended questionnaire:

Domain	Questions
Emotion	How did you feel after using the AI model?
Trust	After seeing (XAI / uncertainty scores / uncertainty score + XAI) produced by the models, I felt...because... How much do you trust the results of the AI model? Was there enough information to help you understand how the AI model made its decisions? Why? Did you understand the AI results? Why? What made it easier for you to understand the AI results? What made it harder for you to understand the AI results?
Self-efficacy	What information from the AI helped you feel more capable of deciding the likelihood in which the patient has colon / lung cancer? Do you think you can effectively use the AI model to help you predict the likelihood in which a patient has colon/lung cancer? Why? Do you feel more or less confident in your clinical judgement when you use this model? Why?
Agreement	The part of the model's results I agree with is...because... The part of the model's results I disagree with is...because...
Decision-making	If you could better understand how the AI works, do you think you would be more able to predict the likelihood that a patient has cancer? Does the (XAI / uncertainty score / uncertainty score + XAI) make you feel more confident in deciding whether a patient likely has cancer compared to if you were never given this information? The AI method(s) impacted my decision in terms of. . .
Suggestions	What do you hope this (XAI / uncertainty score / uncertainty score + XAI) can be used to achieve in the future? It would be more helpful if the (XAI / uncertainty score / uncertainty score + XAI) could... What challenges or limitations do you think the (XAI / uncertainty score / uncertainty score + XAI) may have in its application in the medical field?

A.5 Box and whisker plots of each dimension.



A.5 Descriptive statistics results

Descriptive statistics of the effects of XAI and UQ on Emotion

		Group 1	Group 2	Group 3	Group 4
Positive affect score	Mean (max = 50)	23.11	23.73	27.33	23.43
	Median	21	24	29	22.5
	SD	0.93	0.96	1.02	0.98
	Min	10	10	10	13
	Max	39	40	43	43
Negative affect score	Mean (max = 50)	15.43	13.52	13.97	12.54
	Median	12	12	13	11
	SD	0.50	0.50	0.57	0.33
	Min	10	10	9	10
	Max	34	28	23	34

Descriptive statistics of the effects of XAI and UQ on Trust

		Group 1	Group 2	Group 3	Group 4
Reliability	Mean (max = 15)	6.86	8.24	7.94	8.11
	Median	7	9	8	7.5
	Standard deviation (SD)	1.40	2.36	2.50	2.41
	Min	2	3	3	3
	Max	8	12	13	12
Technical competence	Mean (max = 20)	13.84	13.45	13.76	12.57
	Median	14	14	14	13
	SD	2.37	2.18	2.62	2.70
	Min	8	7	7	6
	Max	18	18	19	17
Understandability	Mean (max = 25)	18.00	18.03	17.48	17.96
	Median	19	18	17	19
	SD	3.63	2.70	3.47	3.11
	Min	7	9	7	11
	Max	22	24	24	25
Faith	Mean (max = 25)	13.51	11.21	12.82	10.46
	Median	14	10	13	11
	Standard deviation (SD)	5.36	3.89	4.43	3.17
	Min	5	5	5	5
	Max	21	21	23	18
Personal attachment	Mean (max = 25)	15.16	13.55	14.70	12.96
	Median	17	14	15	13
	Standard deviation (SD)	4.55	4.45	3.86	4.38
	Min	5	5	5	5
	Max	20	21	25	20
Overall score	Mean (max = 105)	67.38	64.48	66.70	62.07
	Median	68	67	66	63.5
	Standard deviation (SD)	11.67	11.83	13.62	10.65
	Min	47	43	33	35
	Max	85	88	103	78

Descriptive statistics of the effects of XAI and UQ on Usability

		Group 1	Group 2	Group 3	Group 4
Usability	Mean (max = 32)	18.22	20.45	21.11	20.71
	Median	17	21	22	20
	Standard deviation (SD)	3.62	4.98	3.54	5.19
	Min	14	4	12	9
	Max	28	29	29	30